Functional Dependencies and BCNF-

1. Minimal Functional Dependency Set

Definition

A functional dependency means that one attribute uniquely determines another. For example, if we know the *tourist_id*, we can determine the tourist's name, contact details and other related information. A minimal functional dependency set is the smallest set of such dependencies that still represents all the required relationships. In this set, each dependency has only one attribute on the right-hand side, no dependency is redundant and no extra attribute exists on the left hand side.

Conditions for a Minimal FD Set

First, every functional dependency must have only one attribute on the RHS. If an attribute determines multiple attributes, it is written as separate FDs.

Second, no FD in the set should be removable. Each FD must carry unique information that cannot be derived from the others.

Third, no attribute on the left-hand side should be removable. Every attribute in the LHS must be necessary to determine the RHS.

2. Boyce-Codd Normal Form(BCNF)

Definition

A relation is said to be in BCNF if for every functional dependency the LHS is a superkey. A superkey is any attribute or combination of attributes that can uniquely identify each record in a table. In simpler terms, only keys should determine other attributes.

How to Check Whether a Relation is in BCNF

Step 1: Find the candidate key of the relation. This is the attribute or set of attributes that uniquely identifies each row.

Step 2: List all the functional dependencies

of the relation. Step 3: For each FD, check

whether the LHS is a superkey.

Step 4: If the LHS of every FD is a superkey, the relation is in BCNF. If even one FD has an LHS that is not a superkey, the relation violates BCNF and must be decomposed.

In this schema, every relation satisfies BCNF because in each relation the LHS of every FD is the candidate key of that relation, which is always a superkey. There are no partial or transitive dependencies through non-key attributes anywhere in the schema

Minimal FD Set and BCNF Proof

Abstract:
This section outlines the minimal set of Functional Dependencies (FDs) for each relation in the Gujarat Tourism Management System database. Furthermore, it provides a formal proof that all relations satisfy Boyce-Codd Normal Form (BCNF). A relation schema R is in BCNF if, for all non-trivial functional dependencies X → Y in the closure of F (F^+), X is a superkey of R.

1. DISTRICT Relation

Schema:
R(district_id, name, region)

Minimal FD Set (F):

- **fd1: district_id → {name, region}**

Candidate Keys: {district_id}

BCNF Proof:
The only non-trivial functional dependency is fd1. The determinant, district_id, is the primary key and therefore a candidate key. There are no partial or transitive dependencies. Therefore, the DISTRICT relation is in BCNF.

2. TOURIST_SPOT Relation

Schema:
R(spot_id, spot_name, district_id, category, address, established_year, entry_fee_local_inr, entry_fee_domestic_inr, entry_fee_foreign_inr, avg_visit_duration, best_season, is_unesco_heritage, rating, total_reviews, image_url)

Minimal FD Set (F):

- **fd1: spot_id → {spot_name, district_id, category, address, established_year, entry_fee_local_inr, entry_fee_domestic_inr, entry_fee_foreign_inr, avg_visit_duration, best_season, is_unesco_heritage, rating, total_reviews, image_url}**

Candidate Keys: {spot_id}

BCNF Proof:
The only determinant in the minimal FD set is spot_id, which is the primary key of the relation and therefore a superkey. All non-key attributes are directly and fully determined by spot_id.

There are no partial dependencies because the key is a single attribute, and there are no transitive dependencies through non-key attributes. Therefore, TOURIST_SPOT is in BCNF.

3. TOURIST Relation

Schema:
R(tourist_id, first_name, middle_name, last_name, tourist_type, gender, DOB, nationality, phone, email, id_proof_type, id_proof_number)

Minimal FD Set (F):

- **fd1: tourist_id → {first_name, middle_name, last_name, tourist_type, gender, DOB, nationality, phone, email, id_proof_type, id_proof_number}**
- **fd2: phone → {tourist_id}**
- **fd3: email → {tourist_id}**

Candidate Keys: {tourist_id}, {phone}, {email}

BCNF Proof:
The determinants for all non-trivial FDs (fd1, fd2, fd3) are tourist_id, phone, and email respectively. All three determinants are candidate keys (tourist_id as the primary key, phone and email enforced via UNIQUE constraints). Since every determinant is a superkey, the relation is in BCNF.

4. ENTRY_TICKET Relation

Schema:
R(ticket_id, visit_id, tourist_type, valid_from, valid_to, price)

Minimal FD Set (F):

- **fd1: ticket_id → {visit_id, tourist_type, valid_from, valid_to, price}**

Candidate Keys: {ticket_id}

BCNF Proof:
The only determinant is the primary key, ticket_id. All non-key attributes are fully and directly determined by ticket_id alone. There are no partial or transitive dependencies. Therefore, ENTRY_TICKET is in BCNF.

5. EVENT Relation

Schema:

R(event_id, event_name, start_date, end_date, organizer_name, fee_local, fee_domestic, fee_foreign, is_annual, spot_id)

Minimal FD Set (F):

- **fd1: event_id → {event_name, start_date, end_date, organizer_name, fee_local, fee_domestic, fee_foreign, is_annual, spot_id}**

Candidate Keys: {event_id}

BCNF Proof:

The only determinant is the primary key, event_id. Every non-key attribute, including the foreign key spot_id, is fully determined by event_id. There are no partial or transitive dependencies. Therefore, EVENT is in BCNF.

6. SPOT_FACILITY Relation

Schema:

R(spot_id, facility_name, is_available)

Minimal FD Set (F):

- **fd1: {spot_id, facility_name} → {is_available}**

Candidate Keys: {spot_id, facility_name}

BCNF Proof:

The only non-trivial functional dependency is fd1. The determinant {spot_id, facility_name} is the composite primary key and therefore a superkey. The single non-key attribute is_available depends on the full composite key. There are no partial dependencies and no transitive dependencies. Therefore, SPOT_FACILITY is in BCNF.

7. SPOT_TIMING Relation

Schema:

R(spot_id, day_of_week, open_time, close_time, is_closed)

Minimal FD Set (F):

- **fd1: {spot_id, day_of_week} → {open_time}**
- **fd2: {spot_id, day_of_week} → {close_time}**
- **fd3: {spot_id, day_of_week} → {is_closed}**

Candidate Keys: {spot_id, day_of_week}

BCNF Proof:
All three functional dependencies have the composite key {spot_id, day_of_week} as their determinant. Since this composite key is the primary key and therefore a superkey, the left-hand side of every functional dependency is a superkey. Every non-key attribute depends on the full composite key with no partial or transitive dependencies. Therefore, SPOT_TIMING is in BCNF.

8. GUIDE Relation

Schema:
R(guide_id, name, contact_no, is_certified, license_no, years_of_experience, fees_local, fees_domestic, fees_foreign)

Minimal FD Set (F):

- **fd1: guide_id → {name, contact_no, is_certified, license_no, years_of_experience, fees_local, fees_domestic, fees_foreign}**
- **fd2: contact_no → {guide_id}**
- **fd3: license_no → {guide_id}**

Candidate Keys: {guide_id}, {contact_no}, {license_no}

BCNF Proof:
The determinants for all non-trivial FDs are guide_id, contact_no, and license_no. All three are candidate keys (guide_id as primary key, contact_no and license_no enforced as UNIQUE). Since every determinant is a candidate key and therefore a superkey, the relation is in BCNF.

9. GUIDE_LANGUAGE Relation

Schema:
R(guide_id, language)

Minimal FD Set (F):

- **F = ∅ (Trivial dependencies only)**

Candidate Keys: {guide_id, language}

BCNF Proof:
The relation consists entirely of a composite primary key with no non-key attributes. There are no non-trivial functional dependencies. Therefore, GUIDE_LANGUAGE trivially satisfies BCNF.

10. GUIDE_ASSIGNMENT Relation

Schema:
R(spot_id, guide_id, assigned_date, fee_charged)

Minimal FD Set (F):

- **fd1: {spot_id, guide_id, assigned_date} → {fee_charged}**

Candidate Keys: {spot_id, guide_id, assigned_date}

BCNF Proof:
The only non-trivial functional dependency has the full composite primary key {spot_id, guide_id, assigned_date} as its determinant, which is a superkey. The single non-key attribute fee_charged depends on the entire composite key. There are no partial or transitive dependencies. Therefore, GUIDE_ASSIGNMENT is in BCNF.

11. VISIT Relation

Schema:
R(visit_id, visit_date, entry_time, exit_time, ticket_count, tourist_type, total_fee_paid, tourist_id, spot_id)

Minimal FD Set (F):

- **fd1: visit_id → {visit_date, entry_time, exit_time, ticket_count, tourist_type, total_fee_paid, tourist_id, spot_id}**

Candidate Keys: {visit_id}

BCNF Proof:

The only determinant is the primary key, visit_id. All non-key attributes, including the foreign keys tourist_id and spot_id, are fully determined by visit_id. The foreign keys do not themselves determine any other attribute within this relation. There are no partial or transitive dependencies. Therefore, VISIT is in BCNF.

12. FEEDBACK Relation

Schema:

R(feedback_id, visit_id, feedback_type, rating, comments, feedback_date, complain_against, status, resolved_date)

Minimal FD Set (F):

- **fd1: feedback_id → {visit_id, feedback_type, rating, comments, feedback_date, complain_against, status, resolved_date}**

Candidate Keys: {feedback_id}

BCNF Proof:

The only determinant is the primary key, feedback_id. All non-key attributes are fully and directly determined by feedback_id. There are no partial or transitive dependencies anywhere in this relation. Therefore, FEEDBACK is in BCNF.

13. TOURIST_PACKAGE Relation

Schema:

R(package_id, package_name, operator_name, package_type, duration_days, max_group_size, price_local, price_domestic, price_foreign, start_date, end_date, is_transport_included, is_accommodation_included, is_guide_included)

Minimal FD Set (F):

- **fd1: package_id → {package_name, operator_name, package_type, duration_days, max_group_size, price_local, price_domestic, price_foreign, start_date, end_date, is_transport_included, is_accommodation_included, is_guide_included}**

Candidate Keys: {package_id}

BCNF Proof:

The only determinant is the primary key, package_id. All non-key attributes are fully determined by package_id alone. There are no partial or transitive dependencies. Therefore, TOURIST_PACKAGE is in BCNF.

14. PACKAGE_SPOT Relation

Schema:
R(package_id, spot_id, day_number, visit_order_no)

Minimal FD Set (F):

- **fd1: {package_id, spot_id} → {day_number}**
- **fd2: {package_id, spot_id} → {visit_order_no}**

Candidate Keys: {package_id, spot_id}

BCNF Proof:
Both functional dependencies have the composite primary key {package_id, spot_id} as their determinant, which is a superkey. All non-key attributes depend on the full composite key with no partial or transitive dependencies. Therefore, PACKAGE_SPOT is in BCNF.

15. PACKAGE_BOOKING Relation

Schema:
R(booking_id, package_id, tourist_id, booking_date, tour_start_date, group_size, advance_payment, booking_status, amount_to_be_paid, payment_status)

Minimal FD Set (F):

- **fd1: booking_id → {package_id, tourist_id, booking_date, tour_start_date, group_size, advance_payment, booking_status, amount_to_be_paid, payment_status}**

Candidate Keys: {booking_id}

BCNF Proof:
The only determinant is the primary key, booking_id. All non-key attributes, including the foreign keys package_id and tourist_id, are fully determined by booking_id and do not create any transitive dependency within this relation. Therefore, PACKAGE_BOOKING is in BCNF.

16. ACCOMMODATION Relation

Schema:
R(accommodation_id, name, type, address, contact_no, total_rooms, available_rooms, rating, price_local, price_domestic, price_foreign, website_url, is_govt_approved, district_id)

Minimal FD Set (F):

- **fd1: accommodation_id → {name, type, address, contact_no, total_rooms, available_rooms, rating, price_local, price_domestic, price_foreign, website_url, is_govt_approved, district_id}**

Candidate Keys: {accommodation_id}

BCNF Proof:

The only determinant is the primary key, accommodation_id. All non-key attributes, including the foreign key district_id, are fully and directly determined by accommodation_id. There are no partial or transitive dependencies. Therefore, ACCOMMODATION is in BCNF.

17. ACCOMMODATION_ALLOTMENT Relation

Schema:

R(accommodation_id, package_id, check_in_date, check_out_date)

Minimal FD Set (F):

- **fd1: {accommodation_id, package_id} → {check_in_date}**
- **fd2: {accommodation_id, package_id} → {check_out_date}**

Candidate Keys: {accommodation_id, package_id}

BCNF Proof:

Both functional dependencies have the composite primary key {accommodation_id, package_id} as their determinant, which is a superkey. Both non-key attributes depend on the full composite key with no partial or transitive dependencies. Therefore, ACCOMMODATION_ALLOTMENT is in BCNF.

18. TRANSPORT_ROUTE Relation

Schema:

R(route_id, source_district_id, destination_district_id, transport_mode, distance_km, avg_duration_hrs)

Minimal FD Set (F):

- **fd1: route_id → {source_district_id, destination_district_id, transport_mode, distance_km, avg_duration_hrs}**

Candidate Keys: {route_id}

BCNF Proof:

The only determinant is the primary key, route_id. All non-key attributes are fully determined by route_id alone. There are no partial or transitive dependencies. Therefore, TRANSPORT_ROUTE is in BCNF.